

Are galaxies in compact groups special?

Matthieu Tricottet¹ *, Gary A. Mamon², and Eugenia Díaz-Giménez^{3,4}

¹ 80, rue d'Alésia, 75014 Paris, France

² Institut d'Astrophysique de Paris (UMR 7095: CNRS & Sorbonne Université), 98 bis boulevard Arago, 75014 Paris, France

³ CONICET. Instituto de Astronomía Teórica y Experimental (IATE), Laprida 854, X5000BGR, Córdoba, Argentina

⁴ Universidad Nacional de Córdoba (UNC). Observatorio Astronómico de Córdoba (OAC), Laprida 854, X5000BGR, Córdoba, Argentina

Received XXX; accepted YYY

ABSTRACT

We investigate the properties of galaxies in compact groups (CGs) and compare them to a control sample of galaxies.

Key words. galaxies: clusters: general – catalogs

1. Introduction

2. Data

2.1. Samples

We use the compact groups and control samples we built in our previous article [Tricottet et al. \(2025\)](#). We nevertheless briefly recall here how they were built. [Explain.](#) Contrarily to our previous article, though, we remove CGs that we classified as *split* following [Zheng & Shen \(2021\)](#). In this process, our initial compact group sample of 781 groups reduces to 62. We also remove groups built from Lim group 3688 from our Control_{4B} and Control_{4C} samples. This group has six galaxies having a redshift between 0.032 and 0.034, and one having a redshift of 0.048. This group seems spurious, and the outlier galaxy, although not being the BGG, is selected both in our original Control_{4B} and Control_{4C}, influencing the velocity dispersion of the group significantly (1979 and 1981 km s⁻¹, respectively).

We use the SDSS DR 16 to retrieve masses, star formation rates (SFRs) and morphologies of galaxies assessed in Galaxy Zoo ([Lintott et al. \(2011\)](#)). We specifically extracted fields `sfr_tot_p50`, `specsfr_tot_p50` and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table.

2.2. sSFR

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 through the query displayed in listing 1.

```
p.petroMag_r - p.extinction_r as
r_obs,
p.petroMag_u - p.extinction_u as
u_obs,
p.petroMag_g - p.extinction_g as
g_obs,
p.petroMag_i - p.extinction_i as
i_obs,
p.petroMag_z - p.extinction_z as
z_obs,
p.objID,
g.sfr_tot_p50, g.specsfr_tot_p50, g.
lgm_tot_p50,
1.h_alpha_eqw, 1.h_beta_eqw, 1.
oiii_5007_eqw, 1.nii_6584_eqw,
1.h_alpha_flux, 1.h_beta_flux, 1.
oiii_5007_flux, 1.nii_6584_flux,
z.p_el_debiased AS p_E,
z.p_cs_debiased AS p_S
FROM SpecObj AS s
JOIN PhotoObj AS p ON s.bestObjID = p
.objID
JOIN galSpecExtra as g ON s.specObjID
= g.specObjID
JOIN galSpecLine as l ON s.specObjID
= l.specObjID
JOIN zooSpec AS z ON s.specObjID = z.
specObjID
WHERE s.z BETWEEN 0.005 AND 0.0452
AND (p.petroMag_r - p.extinction_r <=
17.77)
AND s.class = 'GALAXY'
AND g.lgm_tot_p50 > -1000
```

Listing 1: Query used for selecting spectral & photometric data from the SDSS

```
SELECT
s.specObjID,
s.z,
```

We select non-AGN galaxies by first placing them on the classical BPT diagnostic of [Veilleux & Osterbrock \(1987\)](#) and requiring measured values of both emission-line ratios

$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha)$ and $\log_{10}([\text{O III}]\lambda 5007/\text{H}\beta)$.

* matthieu.tricottet@gmail.com

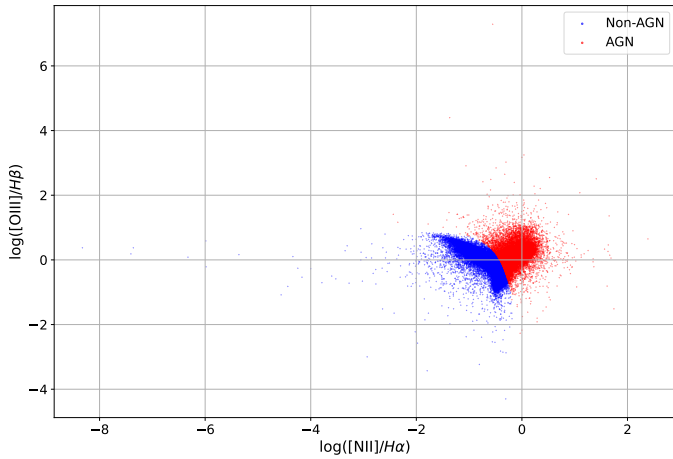


Fig. 1: BPT diagram showing our separation between ordinary and AGN galaxies

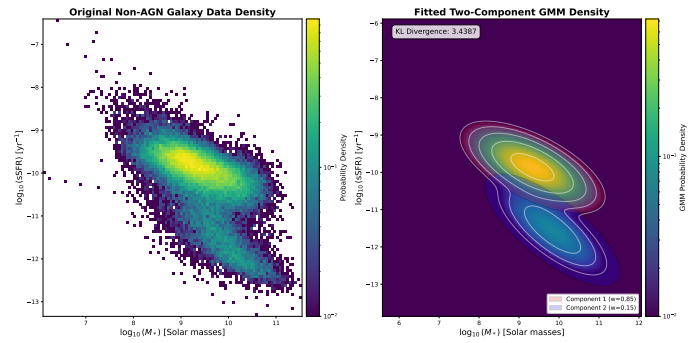


Fig. 2

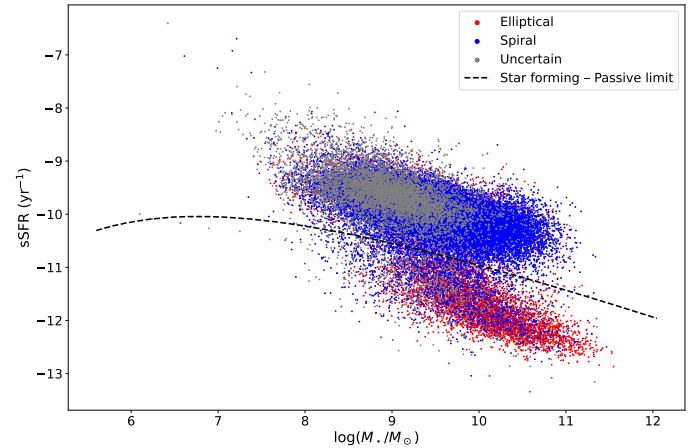


Fig. 3: sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

A galaxy is flagged as an AGN if it satisfies either

$$\log_{10} \frac{[\text{NII}]}{\text{H}\alpha} > 0,$$

or if it lies above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10} \frac{[\text{OIII}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{NII}]/\text{H}\alpha) - 0.05} + 1.3,$$

in which case it is removed from the non-AGN sample. All remaining galaxies—including those with missing line ratios—are retained as non-AGN. The non-AGN selection process is shown on figure 1.

We identify the galaxies having a `specsf_tot_p50` of -9999 in the SDSS as quenched (for graphic representation purposes, we give it here an arbitrary low value of $10^{-14.0} \text{ Gyr}^{-1}$). We then consider the other ones and sort them between the star-forming and passive populations using a two-component Gaussian Mixture Model (GMM) in the $\log M_*$ – $\log \text{sSFR}$ plane. Its parameters determined by minimizing the Kullback–Leibler divergence between the empirical and model densities (Kullback & Leibler (1951); McLachlan & Peel (2000)). For any trial parameter vector θ , we reconstruct the mixture means $\{\mu_i\}$, covariances $\{\Sigma_i\}$, and weights $\{w_i\}$ via a Cholesky-like decomposition and logistic mapping, optionally constraining the second component’s mean to isolate the passive population (Dempster et al. (1977)). The KL divergence is estimated by constructing a 2D histogram of the data and summing $p_{\text{data}} \ln(p_{\text{data}}/p_{\text{GMM}})$, with a small regularization ϵ to avoid singularities (Tojeiro et al. (2009); Bisigello et al. (2018)). We optimize θ via L-BFGS-B (with fallback to Nelder–Mead) over multiple guided and random initializations, selecting the solution with lowest divergence (Pedregosa et al. (2011); Bishop (2006)). Initial means are seeded by a median split in $\log \text{sSFR}$ to improve convergence and robustness against local minima (Rousseeuw (1987)). In the final fit, one component naturally aligns with the high-sSFR “star-forming sequence,” while the other captures the low-sSFR “passive” cloud, providing a data-driven division that agrees with previous multi-Gaussian decompositions of the $\text{SFR} - M_*$ plane (Wuyts et al. (2011); Hahn et al. (2019)). The decision boundaries, defined as the

loci where the posterior probabilities of adjacent components are equal, provide an objective criterion for delineating star-forming from passive galaxies.

Add relevant GM parameters found by the algorithm.

The sSFR status for each sample is shown in table 1. There is no significant differences of for Star forming proportion between CG_4 and the control samples (probabilities estimated using Fisher exact test are 6.5×10^{-1} versus Control_{4B} sample, 2.5×10^{-1} versus Control_{4C} and 1.8×10^{-2} versus RG₄).

Restrictions of the comparison to BGGs and satellites only are also shown in table 1. The p-values found from two-sided Fisher exact test are 0.58 against Control_{4B}, 0.59 against Control_{4C} and 0.37 against RG₄. For satellites, the corresponding p-values are 0.30 against Control_{4B}, 0.03 against Control_{4C} and 0.00 against RG₄.

2.3. Morphology

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions Lintott et al. (2011). Those fractions are provided in the `zooSpecz` table of the SDSS databasethrough the `p_el_debiased` and `p_cs_debiased` columns. We attribute to galaxies the morphology “elliptical” if the `p_el_debiased` value is greater than 0.5, and “spiral” if the `p_cs_debiased` value is greater

Sample		Quenched	Passive	Starforming
CG ₄		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	<i>BGG</i>	6 (9.7 %)	45 (72.6 %)	11 (17.7 %)
	<i>Satellites</i>	12 (6.5 %)	101 (54.3 %)	73 (39.2 %)
Control _{4B}		181 (6.5 %)	1596 (57.1 %)	1019 (36.4 %)
	<i>BGG</i>	58 (8.3 %)	533 (76.4 %)	107 (15.3 %)
	<i>Satellites</i>	122 (5.8 %)	1061 (50.7 %)	911 (43.5 %)
Control _{4C}		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	<i>BGG</i>	65 (8.7 %)	568 (75.6 %)	118 (15.7 %)
	<i>Satellites</i>	125 (5.5 %)	1050 (46.6 %)	1078 (47.8 %)
RG ₄		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	<i>BGG</i>	3 (5.4 %)	38 (67.9 %)	15 (26.8 %)
	<i>Satellites</i>	2 (1.2 %)	63 (37.5 %)	103 (61.3 %)
<i>SDSS selection</i>		634 (1.2 %)	7705 (14.7 %)	44192 (84.1 %)

Table 1: Number of galaxies in each sSFR status for each sample.

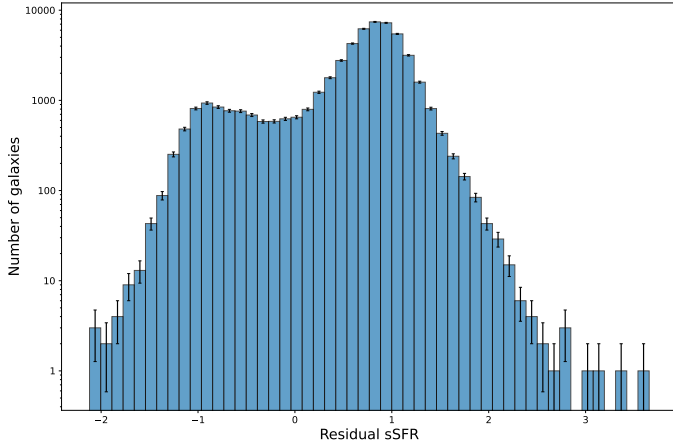
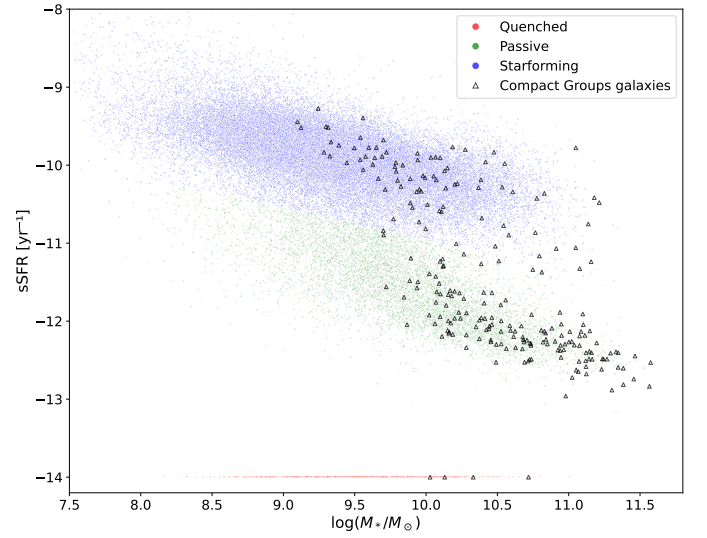


Fig. 4


 Fig. 5: sSFR and morphologies of galaxies in CG₄ and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0} \text{ Gyr}^{-1}$.

than 0.5. Other cases are considered as “uncertain”. The morphologies found for each sample are shown in table 2. Barnard two-sided exact tests were performed to compare the fraction of each morphology between CG₄ and each control sample. The p-values found are 1.4×10^{-3} for ellipticals and 1.2×10^{-3} for spirals versus Control_{4B} sample, 3.3×10^{-3} and 3.4×10^{-2} versus Control_{4C} and 2.2×10^{-7} and 5.9×10^{-7} versus RG₄. Morphologies of CG₄, which we identified as cores of richer groups in Tricottet et al. (2025), have morphologies significantly different from ordinary groups of four members. Figure 5 displays the sSFR against masses for galaxies from the SDSS selection and for CG₄s galaxies.

2.4. Morphology vs sSFR

Starforming CG₄ are composed of 58 of their galaxies classified as spiral (84.1% once galaxies having uncertain morphology removed) and 11 (15.9% once galaxies having uncertain morphology removed) elliptical. These figures are 809 (88.4%) spiral and 106 (11.6%) elliptical galaxies for Control_{4B}, 851 (84.3%) spiral and 159 (15.7%) elliptical galaxies for Control_{4C} and 91 (91%) spiral and 9 (9%) elliptical galaxies for RG₄. The probability that those fractions come from the same distribution is estimated from the Barnard two-sided exact test as 0.29 against Control_{4B}, 0.97 against Control_{4C} and 0.19 against RG₄. There is thus

neither significant excess nor lack of starforming elliptical galaxies in CG₄ compared to all our control samples.

If we restrict the comparison to BGGs only, we find 36 Elliptical BGGs (58.1% of the BGGs) and 17 Spiral BGGs (27.4%) in CG₄. This is to be compared to 405 (58.0%) Elliptical and 195 (27.9%) Spiral galaxies in Control_{4B}, 424 (56.5%) Elliptical and 216 (28.8%) Spiral in Control_{4C}, and 21 (37.5%) Elliptical and 26 (46.4%) Spiral in RG₄. The p-values found from two-sided Fisher exact test are 1.00 against Control_{4B}, 0.88 against Control_{4C} and 0.03 against RG₄. We can therefore see that BGG morphologies are different between CG₄ and RG₄: CG₄ BGGs are more likely to be elliptical than RG₄ BGGs.

3. sSFR

3.1. Main sequence

We model the star-forming main sequence as a power law in stellar mass, to various orders (1 to 4) - cf figure 9. We then choose to use order 2 to compute the residuals of each

Sample	Elliptical	Spiral	Uncertain
CG ₄	124 (50.0 %)	86 (34.7 %)	38 (15.3 %)
Control _{4B}	1106 (39.6 %)	1266 (45.3 %)	420 (15.0 %)
Control _{4C}	1215 (40.4 %)	1249 (41.6 %)	540 (18.0 %)
RG ₄	60 (26.8 %)	129 (57.6 %)	35 (15.6 %)
<i>SDSS selection</i>	<i>10091 (19.2 %)</i>	<i>34715 (66.1 %)</i>	<i>7725 (14.7 %)</i>

Table 2: Number of galaxies of each morphology for each sample.

Sample	$\langle \log sSFR \rangle$		$\langle M_r \rangle$		$\langle \log(M_\star/M_\odot) \rangle$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG ₄	-12.08	-10.19	-21.16	-20.4	10.64	10.05
Control _{4B}	-12.1	-10.31	-21.52	-20.75	10.79	10.24
Control _{4C}	-11.92	-10.16	-20.81	-19.93	10.47	9.82
RG ₄	-11.91	-10.16	-21.11	-20.32	10.62	9.97

Table 3: Median values of galaxy properties for passive and star-forming populations in each sample. Error bars are computed via bootstrapping (10 000 iterations).

Sample	sSFR (yr ⁻¹)		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG ₄	-11.53 ± 0.09	-10.85 ± 0.30	-20.93 ± 0.11	-20.77 ± 0.32	10.46 ± 0.06	10.32 ± 0.12
Control _{4B}	-11.53 ± 0.03	-10.86 ± 0.06	-21.25 ± 0.03	-21.12 ± 0.07	10.61 ± 0.02	10.46 ± 0.04
Control _{4C}	-11.3 ± 0.03	-10.62 ± 0.06	-20.42 ± 0.05	-20.55 ± 0.08	10.22 ± 0.02	10.15 ± 0.06
RG ₄	-11.14 ± 0.19	-10.58 ± 0.16	-20.69 ± 0.14	-20.74 ± 0.28	10.32 ± 0.07	10.19 ± 0.12

Table 4: Median values of various quantities for galaxies according to the fertility of their BGG (BGG included). Error bars are computed via bootstrapping (10 000 iterations).

180 star-forming galaxy to the main sequence, and plot their
histograms in figure 10.

The median main-sequence residual of CG₄ galaxies is:

- significantly different from that of Control_{4B}, with a median offset of $\Delta = 0.056$ ($0.028 \leq \Delta \leq 0.098$, bootstrap), corresponding to a p -value of $p = 0.053$.
- significantly different from that of Control_{4C}, with a median offset of $\Delta = 0.082$ ($0.062 \leq \Delta \leq 0.121$, bootstrap), corresponding to a p -value of $p = 0.011$.
- not significantly different from that of RG₄, with a median offset of $\Delta = 0.046$ ($-0.035 \leq \Delta \leq 0.098$, bootstrap), corresponding to a p -value of $p = 0.520$.

A direct rank-sum comparison of the star-forming main-sequence offsets only identifies significant differences between CG₄ and

Control_{4C}, whose median offset is -0.060 with respect to the fitted sequence ($p = 0.03$).

Within CG₄, the star-forming main-sequence offsets of the Embedded, Isolated and Predom classes remain statistically consistent with one another.

200 3.2. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.24$ for 177 satellites

($p = 1.2 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 4.1 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.08$ for 1991 satellites ($p = 3.6 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.3 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.25$ for 2143 satellites ($p = < 10^{-6}$) and Pearson coefficient $r = 0.20$ ($p = < 10^{-6}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.10$ for 2143 satellites ($p = 4.6 \times 10^{-6}$) and Pearson coefficient $r = 0.09$ ($p = 1.8 \times 10^{-5}$).

3.3. Dominance

We split groups into two categories: dominated groups, for which the BGG luminosity fraction exceeds our threshold of 0.6, and non-dominated groups. Among the group properties explored in this work, the quantities showing at least one significant dominated/non-dominated difference are $\langle R_{ij} \rangle$, $\Delta_{\text{BGG-cen}}/\langle R_{ij} \rangle$, σ_v , $\Delta V_{\text{BGG}}/\sigma_v$ and L_{group} .

In CG₄, no dominated/non-dominated difference remains significant after the FDR correction.

In Control_{4B}, the strongest corrected dominated/non-dominated separation concerns L_{group} : the dominated groups have a median value of $9.1 \times 10^{10} L_\odot$, compared to $1.28 \times 10^{11} L_\odot$ for the non-dominated ones ($p_{\text{adj}} = < 10^{-6}$; Cliff's $\delta = -0.55$).

Sample	sSFR (yr^{-1})		M_r	Star-forming BGG	$\log(M_*/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	-11.23 ± 0.16	-10.92 ± 0.48	-20.45 ± 0.12	-20.34 ± 0.18	10.22 ± 0.05	10.09 ± 0.07
Control4B	-11.26 ± 0.06	-10.91 ± 0.13	-20.9 ± 0.03	-20.75 ± 0.06	10.43 ± 0.02	10.3 ± 0.03
Control4C	-10.96 ± 0.04	-10.6 ± 0.07	-19.8 ± 0.04	-20.02 ± 0.09	9.91 ± 0.02	9.89 ± 0.07
RG4	-10.82 ± 0.18	-10.61 ± 0.22	-20.23 ± 0.17	-20.3 ± 0.13	10.09 ± 0.08	10 ± 0.13

Table 5: Median values of various quantities for galaxies according to the fertility of their BGG (satellites only). Error bars are computed via bootstrapping (10 000 iterations).

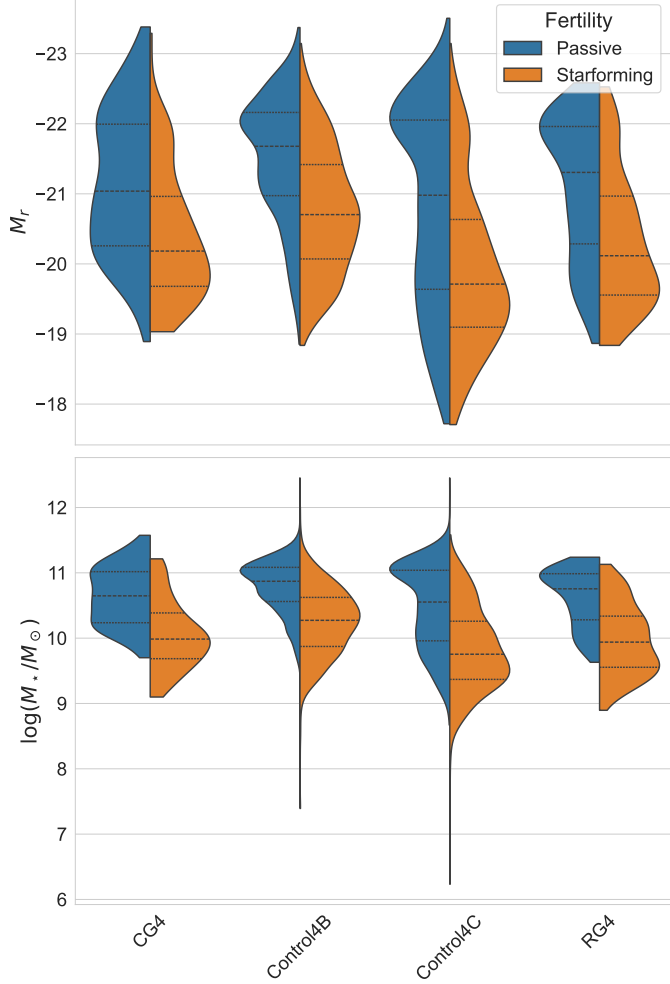


Fig. 6: Distribution various quantities for galaxies according their fertility.

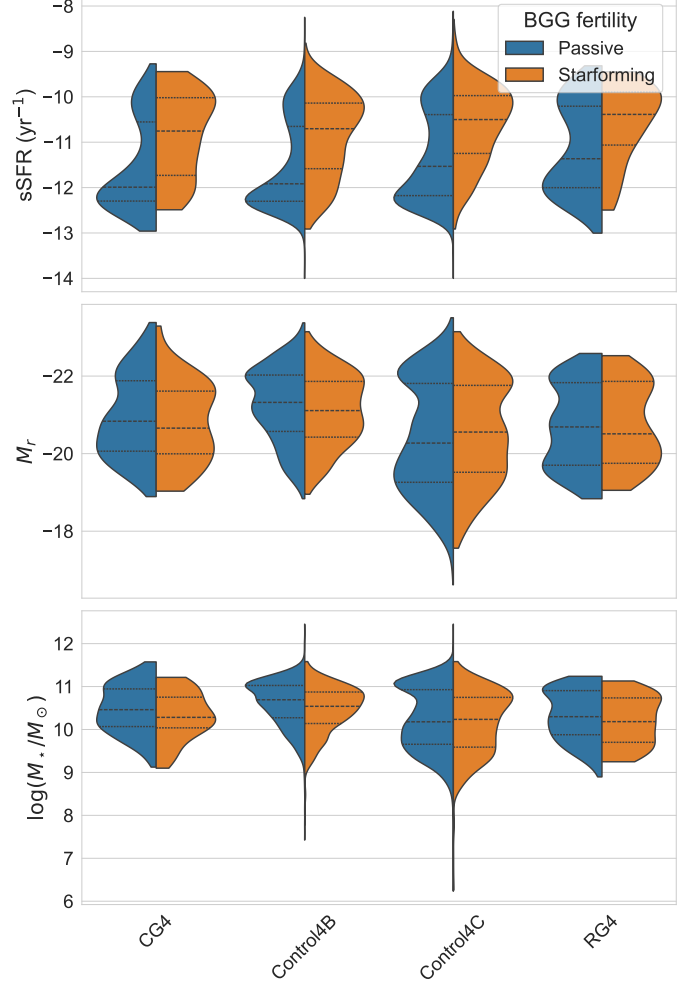


Fig. 7: Distribution various quantities for galaxies according to the fertility of their BGG (BGG included).

In Control_{4C}, the strongest corrected dominated/non-dominated separation concerns L_{group} : the dominated groups have a median value of $8.49 \times 10^{10} L_\odot$, compared to $9.54 \times 10^{10} L_\odot$ for the non-dominated ones ($p_{\text{adj}} = < 10^{-6}$; Cliff's $\delta = -0.25$).

In RG₄, the strongest corrected dominated/non-dominated separation concerns L_{group} : the dominated groups have a median value of $7.09 \times 10^{10} L_\odot$, compared to $9.82 \times 10^{10} L_\odot$ for the non-dominated ones ($p_{\text{adj}} = 7.8 \times 10^{-5}$; Cliff's $\delta = -0.59$).

The group spiral fraction also correlates significantly with the group-scale observables in a subset of the samples.

For Control_{4C} dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ ($\rho_s = 0.38$, 527 groups, $p = < 10^{-6}$).

For Control_{4C} non-dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ ($\rho_s = 0.35$, 224 groups, $p = < 10^{-6}$).

For Control_{4C} dominated groups, S_{frac} correlates with σ_v ($\rho_s = -0.23$, 527 groups, $p = < 10^{-6}$).

For Control_{4C} dominated groups, S_{frac} correlates with L_{group} ($\rho_s = -0.15$, 527 groups, $p = 8.4 \times 10^{-4}$).

For Control_{4B} non-dominated groups, S_{frac} correlates with L_{group} ($\rho_s = -0.14$, 504 groups, $p = 1.4 \times 10^{-3}$).

For CG₄ dominated groups, S_{frac} correlates with L_{group} ($\rho_s = -0.50$, 36 groups, $p = 1.9 \times 10^{-3}$).

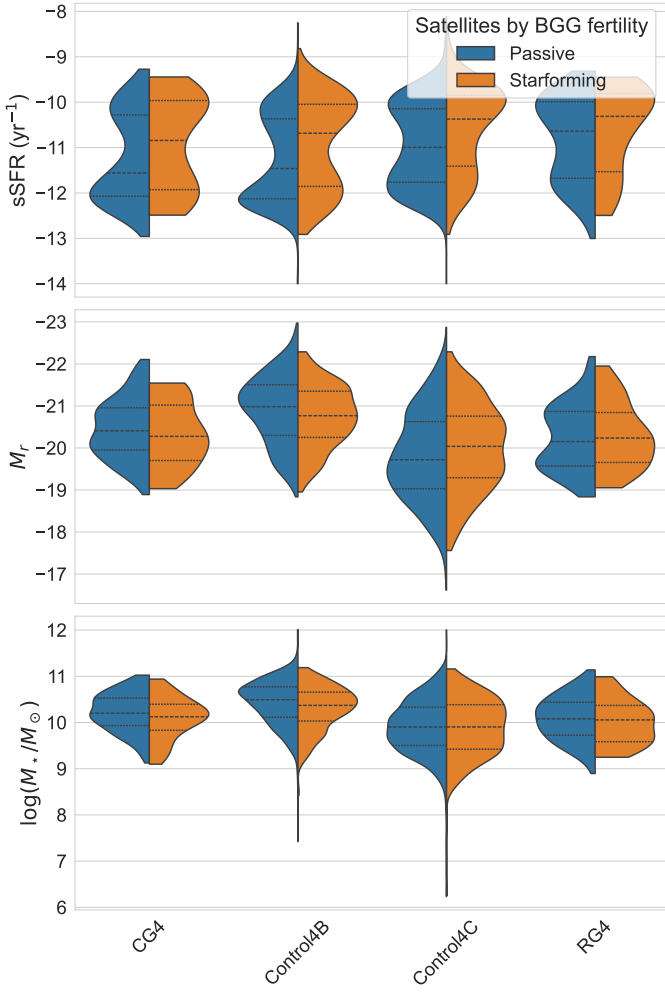


Fig. 8: Distribution various quantities for galaxies according to the fertility of their BGG (BGG excluded).

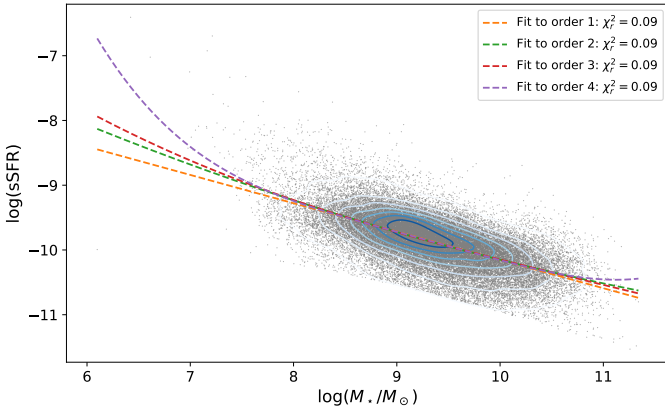


Fig. 9

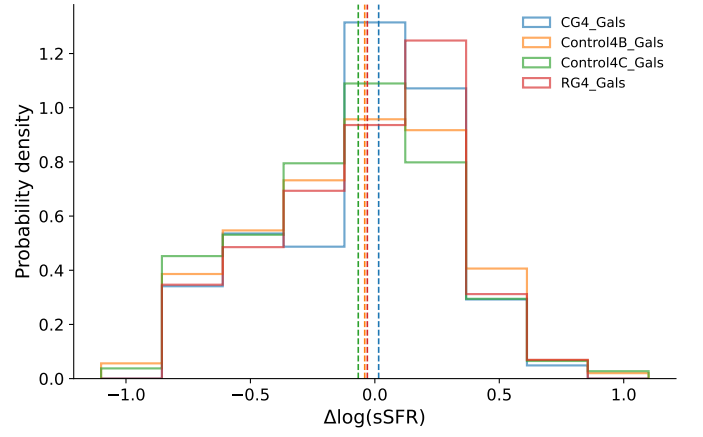


Fig. 10

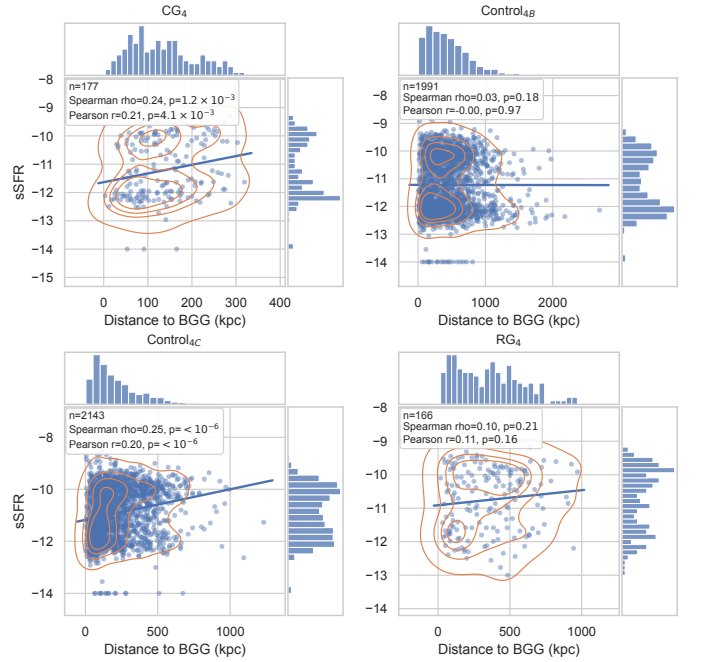


Fig. 11: sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

For Control_{4B} non-dominated groups, S_{frac} correlates with σ_v ($\rho_s = -0.09$, 504 groups, $p = 0.03$).

For Control_{4B} non-dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ ($\rho_s = 0.09$, 504 groups, $p = 0.04$).

For CG₄ non-dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ ($\rho_s = 0.40$, 26 groups, $p = 0.04$).

3.4. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in

Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 ($p = 0.01$), while the satellite-level clustered fit gives an odds ratio of 1.84 ($p = 0.02$);

For Control_{4C} non-dominated groups, S_{frac} correlates with σ_v ($\rho_s = -0.20$, 224 groups, $p = 2.6 \times 10^{-3}$).

For CG₄ non-dominated groups, S_{frac} correlates with $\Delta V_{\text{BGG}}/\sigma_v$ ($\rho_s = -0.53$, 26 groups, $p = 5.6 \times 10^{-3}$).

For Control_{4C} non-dominated groups, S_{frac} correlates with L_{group} ($\rho_s = -0.15$, 224 groups, $p = 0.02$).

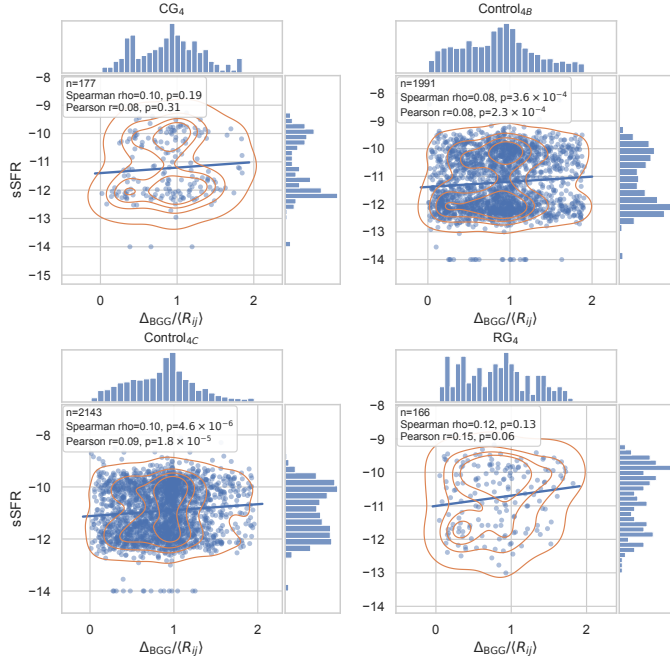


Fig. 12: sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

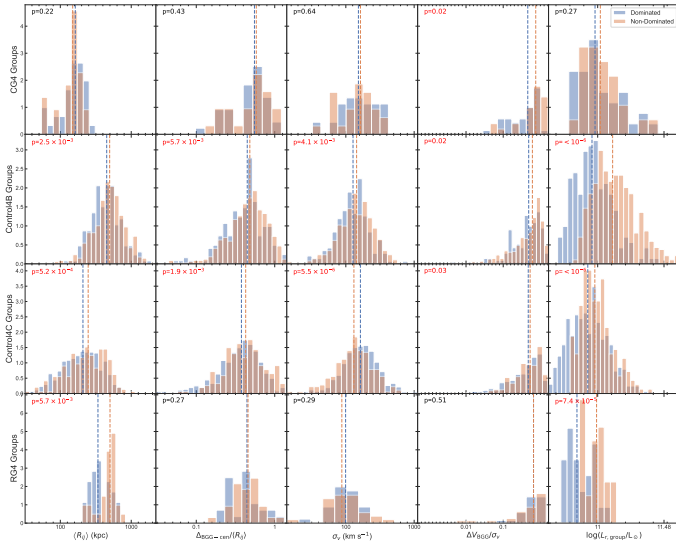


Fig. 13: Distributions of the main group-scale observables in dominated and non-dominated systems.

Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 ($p = 2.1 \times 10^{-5}$), while the satellite-level clustered fit gives an odds ratio of 1.84 ($p = 2.7 \times 10^{-5}$).

4. Group Dynamics

The crossing-time exploration quantifies the trends illustrated in figures 15 to 17.

For $\log(M_{VT}/L_r)$, the global Spearman coefficient with $\log t_{\text{cross}}$ is $\rho_S = -0.39$ based on 816 groups ($p = < 10^{-6}$).

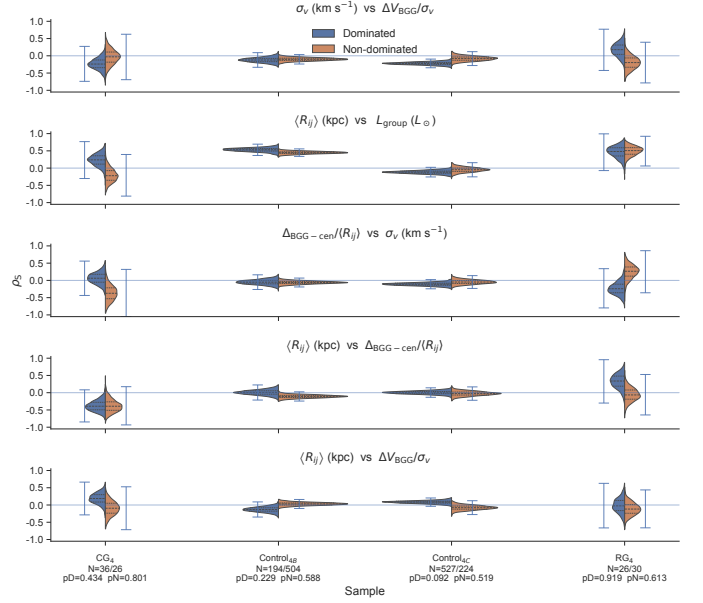


Fig. 14: Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion.

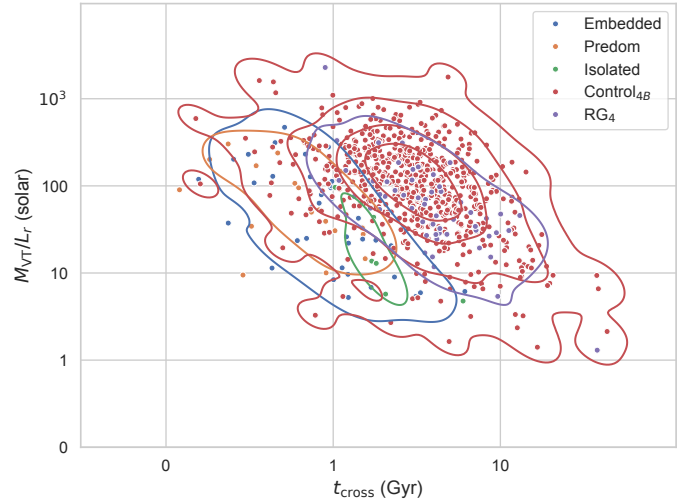


Fig. 15: Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

For $\log(M_{VT})$, the global Spearman coefficient with $\log t_{\text{cross}}$ is $\rho_S = -0.33$ based on 816 groups ($p = < 10^{-6}$).

For $\log(L_{\text{group}})$, the global Spearman coefficient with $\log t_{\text{cross}}$ is $\rho_S = 0.08$ based on 816 groups ($p = 0.03$).

5. Colour

We derive the $(u-r)$, $(u-g)$, $(g-r)$ and $(r-i)$ colours from the matched SDSS photometry. The retained matches are 110 of 248 galaxies in CG₄, 931 of 2792 galaxies in Control_{4B}, 1478 of 3004 galaxies in Control_{4C}, 102 of 224 galaxies in RG₄. Stellar mass is a confounding variable in colour comparisons (Alpaslan et al. 2015); we therefore model each colour jointly with $\log(M_{\star}/M_{\odot})$ rather than comparing unadjusted colour distributions.

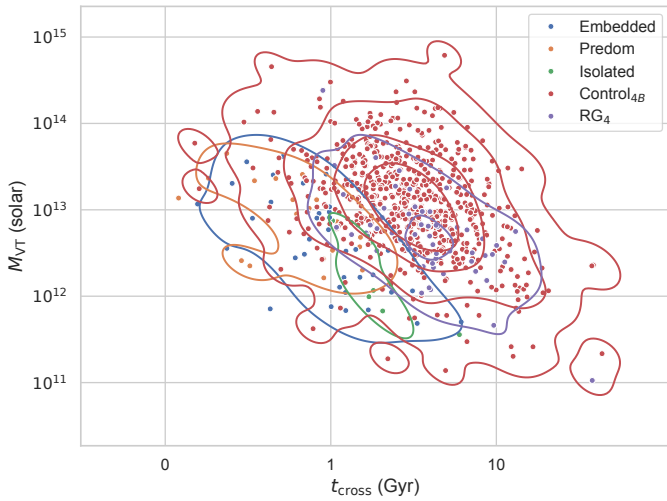


Fig. 16: Crossing time versus virial mass for the compact groups and their controls.

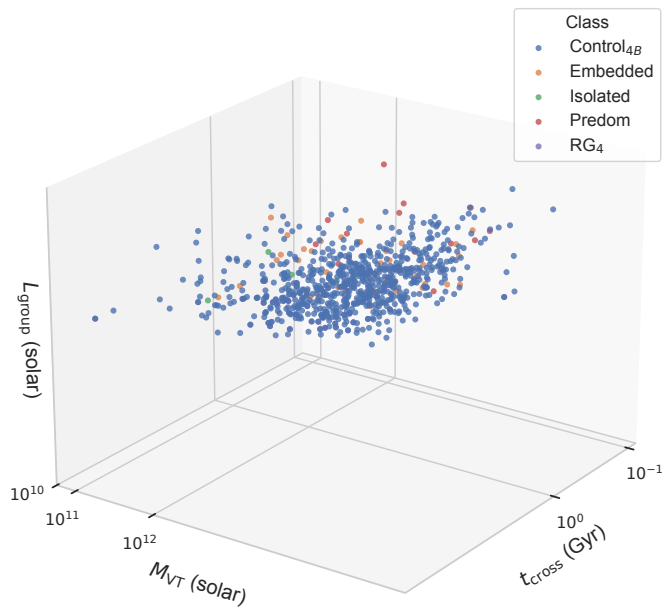


Fig. 17: Three-dimensional view of the crossing time, virial mass and group luminosity space.

5.1. Catalogue colour–mass relations

The hypothesis of a common colour–mass slope across catalogues is rejected for $u-r$ ($p = 4.96 \times 10^{-4}$), $u-g$ ($p = 0.00203$), $g-r$ ($p = 1.46 \times 10^{-4}$), $r-i$ ($p = 0.00495$). After Holm correction of the directional comparisons, significantly steeper CG₄ slopes are found for $u-r$ relative to Control_{4C}, with $\Delta b = 0.289$ ($p_{\text{Holm}} = 0.00163$); for $u-r$ relative to SDSS, with $\Delta b = 0.232$ ($p_{\text{Holm}} = 0.00163$); for $u-g$ relative to Control_{4C}, with $\Delta b = 0.237$ ($p_{\text{Holm}} = 0.00287$); for $u-g$ relative to SDSS, with $\Delta b = 0.186$ ($p_{\text{Holm}} = 0.00165$). These relations are shown in Fig. 18.

5.2. Satellite colours

To isolate an environmental difference, we pool satellites from the control catalogues as the ordinary-group population and compare them with CG₄ satellites at the common reference mass $\log(M_{\star}/M_{\odot}) = 9.885$. At this mass, compact-group satellites are 0.124 mag bluer in $u-r$ ($p_{\text{Holm}} = 0.00909$) and 0.106 mag bluer in $u-g$ ($p_{\text{Holm}} = 0.00737$). The compact and ordinary colour–mass slopes also differ in $u-r$ ($b_{\text{compact}} = 0.819$, $b_{\text{ordinary}} = 0.344$, $p_{\text{Holm}} = 2.21 \times 10^{-5}$), $u-g$ ($b_{\text{compact}} = 0.532$, $b_{\text{ordinary}} = 0.150$, $p_{\text{Holm}} = 1.42 \times 10^{-4}$), $g-r$ ($b_{\text{compact}} = 0.287$, $b_{\text{ordinary}} = 0.195$, $p_{\text{Holm}} = 2.9 \times 10^{-4}$). Figure 19 displays the corresponding fitted relations.

5.3. Robustness of the satellite colour result

The SDSS-photometry subset is not representative of the full galaxy catalogues: matched galaxies are systematically less massive, more actively star-forming and more frequently satellites than unmatched galaxies. The fractions with all four colours are 44.4% in CG₄, 33.3% in Control_{4B}, 49.2% in Control_{4C}, 45.5% in RG₄.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG₄ coefficients is significantly negative after Holm correction. After additionally controlling for morphology and sSFR class, negative coefficients appear in $(u-r)$ ($\Delta = -0.127$ mag, $p_{\text{Holm}} = 0.0177$), $(u-g)$ ($\Delta = -0.101$ mag, $p_{\text{Holm}} = 0.0271$). These adjusted coefficients are control-sample dependent: the significant individual comparisons are Control_{4C} in $(u-r)$, Control_{4C} in $(u-g)$, Control_{4C} in $(g-r)$.

Within sSFR classes, no negative CG₄ residual difference survives correction for the four colours. Likewise, no within-morphology residual difference survives the same correction. Thus, the data do not isolate the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Figure 20 summarizes the model and control-sample dependence. Taken together, these tests do not establish a robust global blue offset for CG₄ satellites; the negative full-model coefficients should instead be treated as a control-sensitive exploratory result.

5.4. Radial and group-level colour trends

The residual colours become bluer at larger projected distance from the BGG in CG₄. After Holm correction within each distance metric, significant CG₄ correlations are found for $(u-r)$ ($\rho_S = -0.379$, $p_{\text{Holm}} = 2.6 \times 10^{-4}$), for $(u-g)$ ($\rho_S = -0.336$, $p_{\text{Holm}} = 0.0011$), for $(g-r)$ ($\rho_S = -0.257$, $p_{\text{Holm}} = 0.00922$). This radial trend is consistent with stronger processing near the group centre and a less processed outer population, rather than centrally triggered blue colours.

At group level, crossing time correlates with mean satellite $\Delta(u-r)$ ($\rho_S = -0.418$, $p_{\text{Holm}} = 0.013$), median satellite $\Delta(u-r)$ ($\rho_S = -0.409$, $p_{\text{Holm}} = 0.0167$), satellite blue fraction ($\rho_S = 0.369$, $p_{\text{Holm}} = 0.0393$). These group-level relations remain exploratory because many groups contribute only a small number of colour-matched satellites.

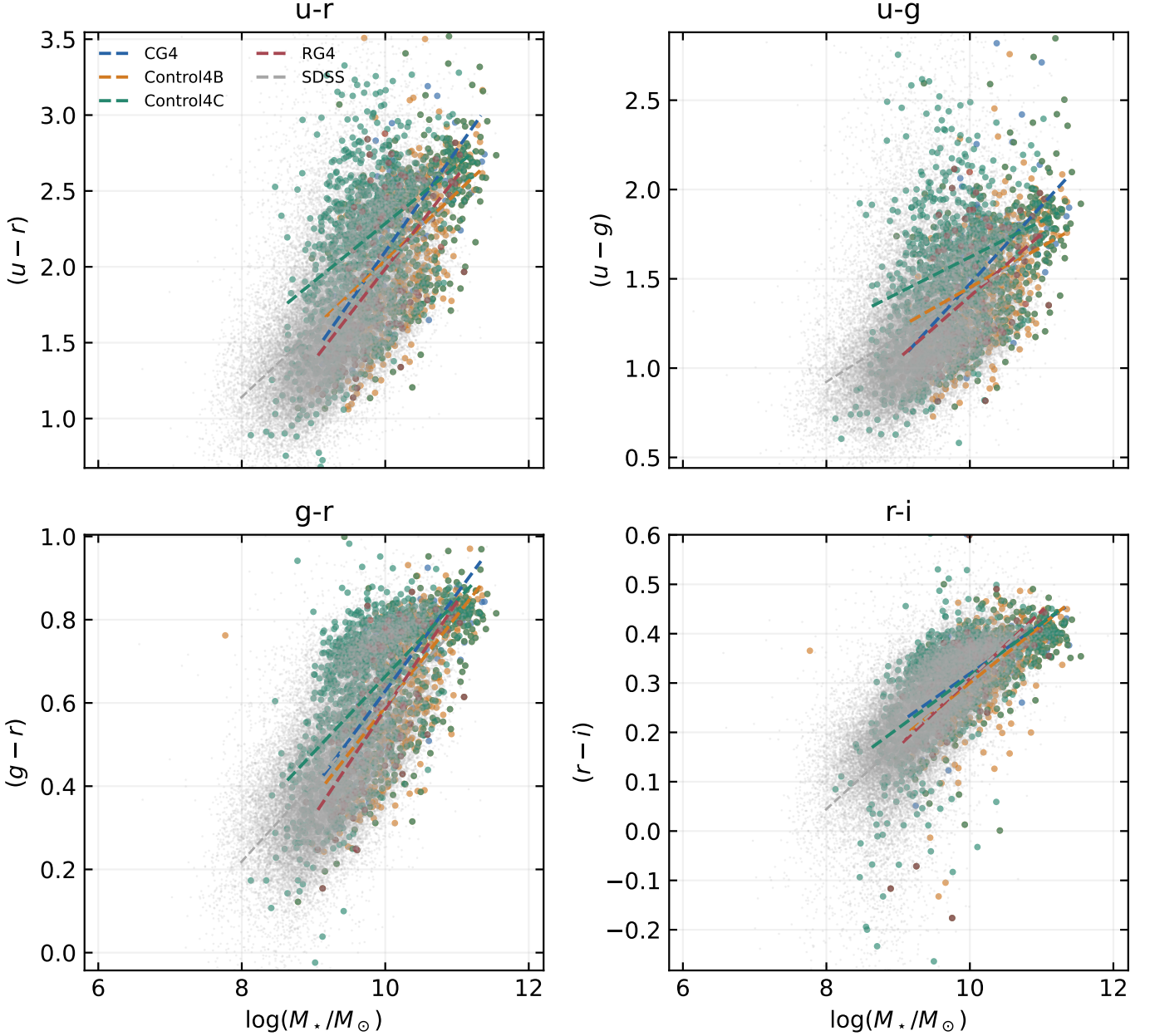


Fig. 18: Optical colour as a function of stellar mass for CG₄, the three control catalogues and the matched SDSS reference sample. Dashed lines show the fitted catalogue relations.

5.5. BGG and satellite colours

The mean satellite colour varies with the BGG colour across the combined group sample: for u-r, $r_P = 0.360$ ($p_{\text{Holm}} = 7.91 \times 10^{-7}$) and $\rho_S = 0.321$ ($p_{\text{Holm}} = 1.3 \times 10^{-5}$); for u-g, $r_P = 0.326$ ($p_{\text{Holm}} = 6.01 \times 10^{-6}$) and $\rho_S = 0.325$ ($p_{\text{Holm}} = 1.3 \times 10^{-5}$); for g-r, $r_P = 0.340$ ($p_{\text{Holm}} = 2.94 \times 10^{-6}$) and $\rho_S = 0.283$ ($p_{\text{Holm}} = 1.13 \times 10^{-4}$); for r-i, $r_P = 0.142$ ($p_{\text{Holm}} = 0.0464$) and $\rho_S = 0.232$ ($p_{\text{Holm}} = 0.00103$). After Holm correction, the BGG–satellite colour slope in CG₄ does not differ significantly from any control catalogue in any of the four colours. Figure 22 shows these group-level relations.

5.6. BGG colours and luminosity domination

The BGG stellar-mass distributions are significantly different between the domination classes ($p = 7.32 \times 10^{-4}$); the median $\log(M_*/M_\odot)$ values are 11.092 for dominated and 10.976 for non-dominated BGGs. Before mass adjustment, a significant colour difference is found in g-r ($\Delta_{\text{median}} = 0.021$, $p_{\text{Holm}} = 0.00157$). At the common reference mass $\log(M_*/M_\odot) = 11.042$, the adjusted results are u-r: $\Delta = -0.002$ mag ($p_{\text{Holm}} = 1$); u-g: $\Delta = -0.010$ mag ($p_{\text{Holm}} = 1$); g-r: $\Delta = 0.007$ mag ($p_{\text{Holm}} = 1$); r-i: $\Delta = 0.001$ mag ($p_{\text{Holm}} = 1$). Thus, no dominated/non-dominated BGG colour offset remains significant after controlling for stellar mass. The mass dependence and class fits are shown in Fig. 23.

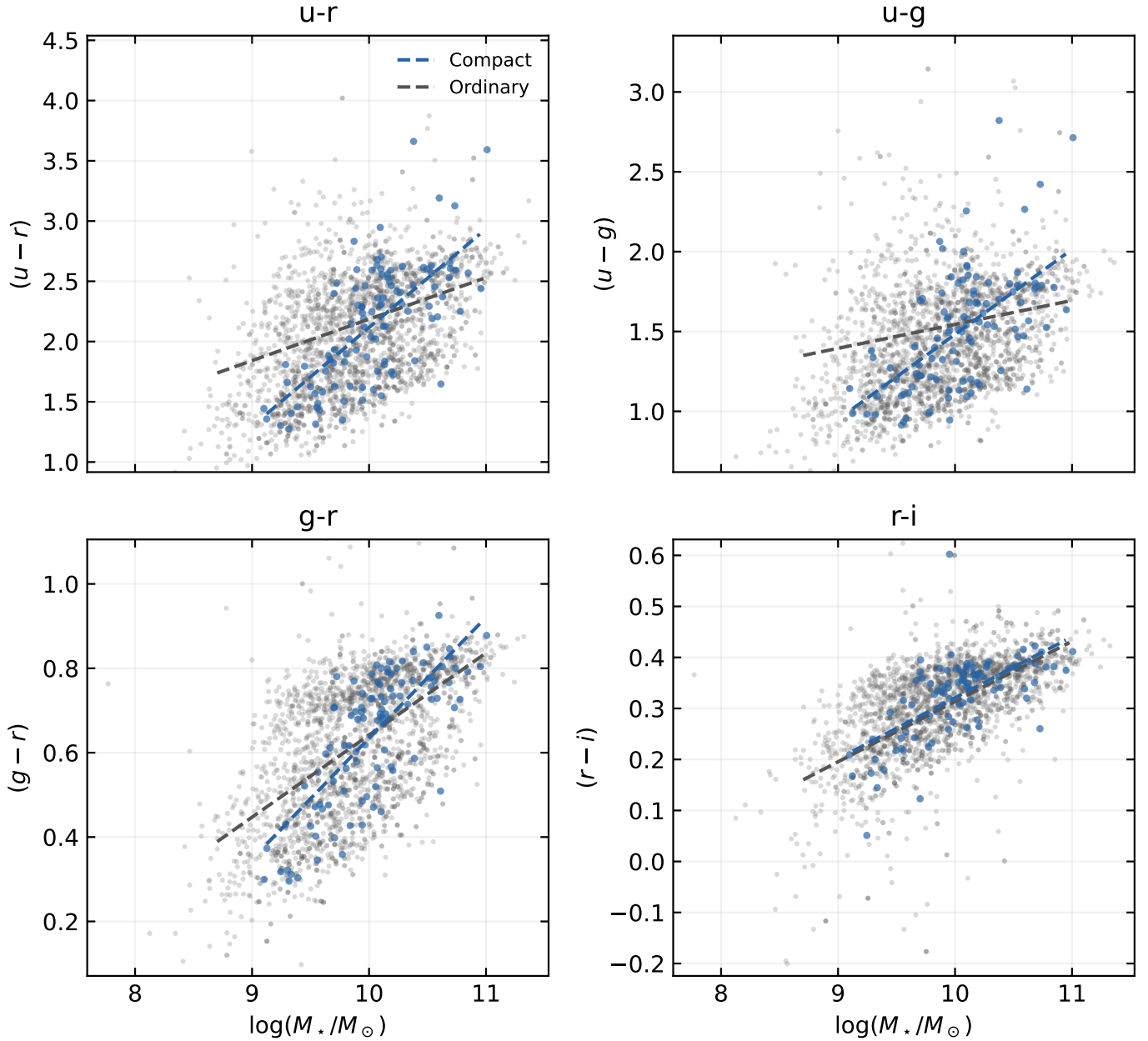


Fig. 19: Satellite colour–mass relations in compact and ordinary groups. The fitted lines expose both colour offsets at the reference mass and differences in slope.

6. Conclusion

After removing systems classified as split, we find that compact groups differ from regular four-member groups mainly through their galaxy population mix. Galaxies in CG4 are more frequently elliptical and less frequently spiral than galaxies in RG4, with highly significant differences in both fractions. This morphological difference is also present among BGGs: CG4 BGGs are more often elliptical than RG4 BGGs.

The star-formation difference is more specific. CG4 satellites are significantly less frequently star-forming than RG4 satellites, while the star-forming fraction of BGGs does not differ significantly between the two samples. Moreover, among galaxies that remain star-forming, CG4 galax-

ies do not show a significant offset from RG4 galaxies relative to the star-forming main sequence.

The colour analysis does not indicate a simple uniform reddening of compact-group galaxies. The pooled mass- and redshift-adjusted comparison does not support a statistically robust blue CG4 satellite offset. Negative coefficients appear after adding morphology and sSFR class, but the result is sensitive to the choice of control catalogue. Neither the sSFR-class nor morphology splits isolate a significant effect among star-forming spirals. The radial residual trends instead indicate that outer CG4 satellites are bluer than inner satellites, consistent with stronger processing toward the BGG. The BGG and mean satellite colours are correlated, but the corresponding CG4 relations do not differ significantly from those of the control catalogues. Like-

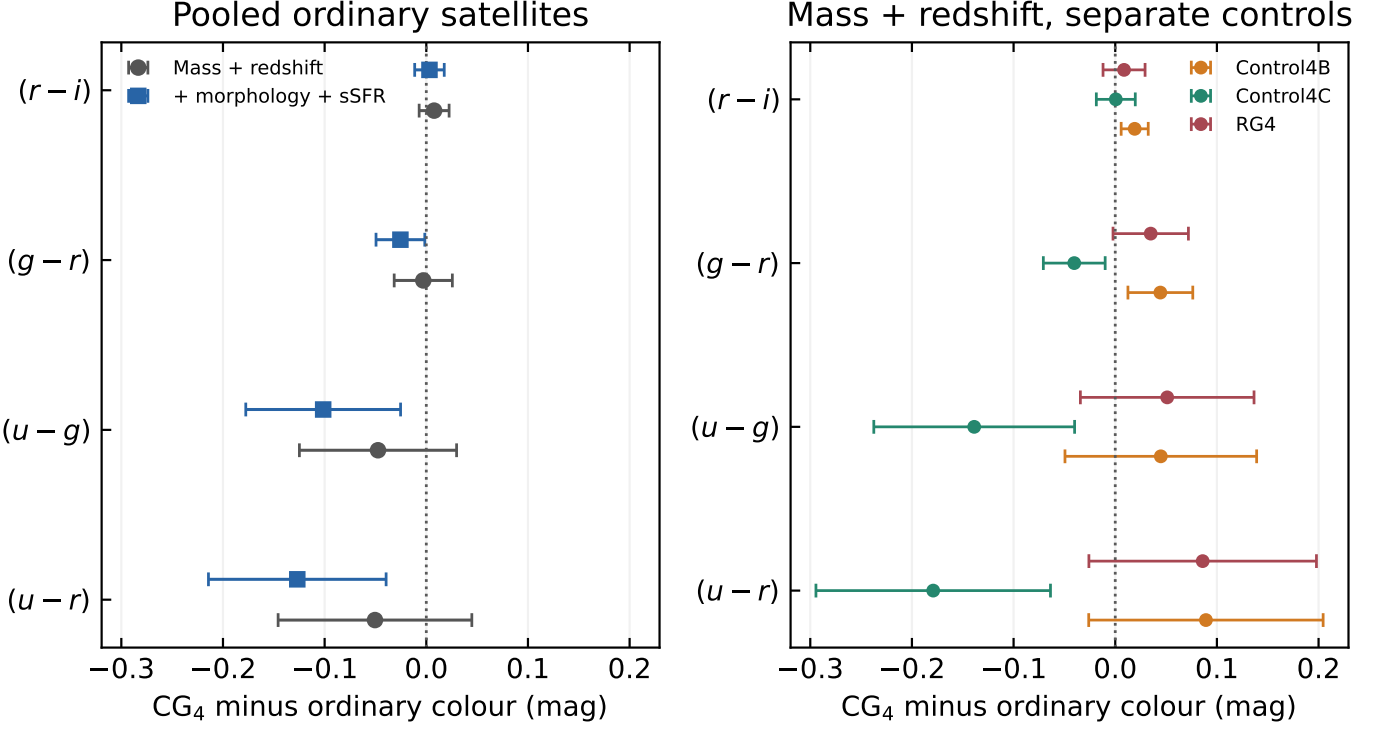


Fig. 20: Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG_4 minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG_4 satellites.

wise, the apparent colour difference between dominated and non-dominated BGGs disappears after controlling for stellar mass.

430 These results suggest that compact groups are not simply denser versions of ordinary four-member groups. Rather, they appear to host a more evolved galaxy population, with an enhanced fraction of early-type galaxies and a reduced fraction of star-forming satellites. The present analysis is therefore consistent with enhanced morphological transformation and quenching in compact-group environments, although it does not yet distinguish between compact-group-specific evolution, pre-processing in larger-scale structures, and selection effects.

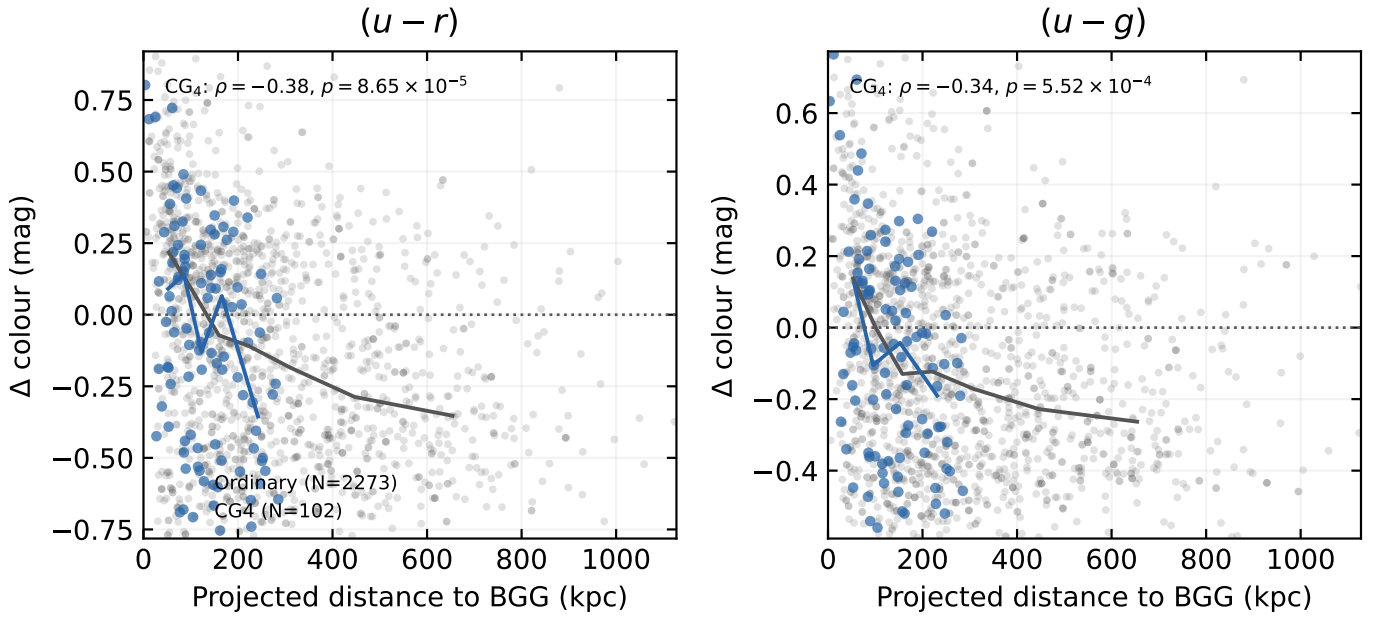


Fig. 21: Mass- and redshift-adjusted satellite $(u-r)$ and $(u-g)$ residuals as a function of projected distance to the BGG. Lines connect binned medians for CG₄ and the pooled ordinary-group satellites; extreme plotting tails are omitted for readability, while the reported rank correlations use the full matched sample. More negative residuals are bluer than the ordinary-satellite reference relation.

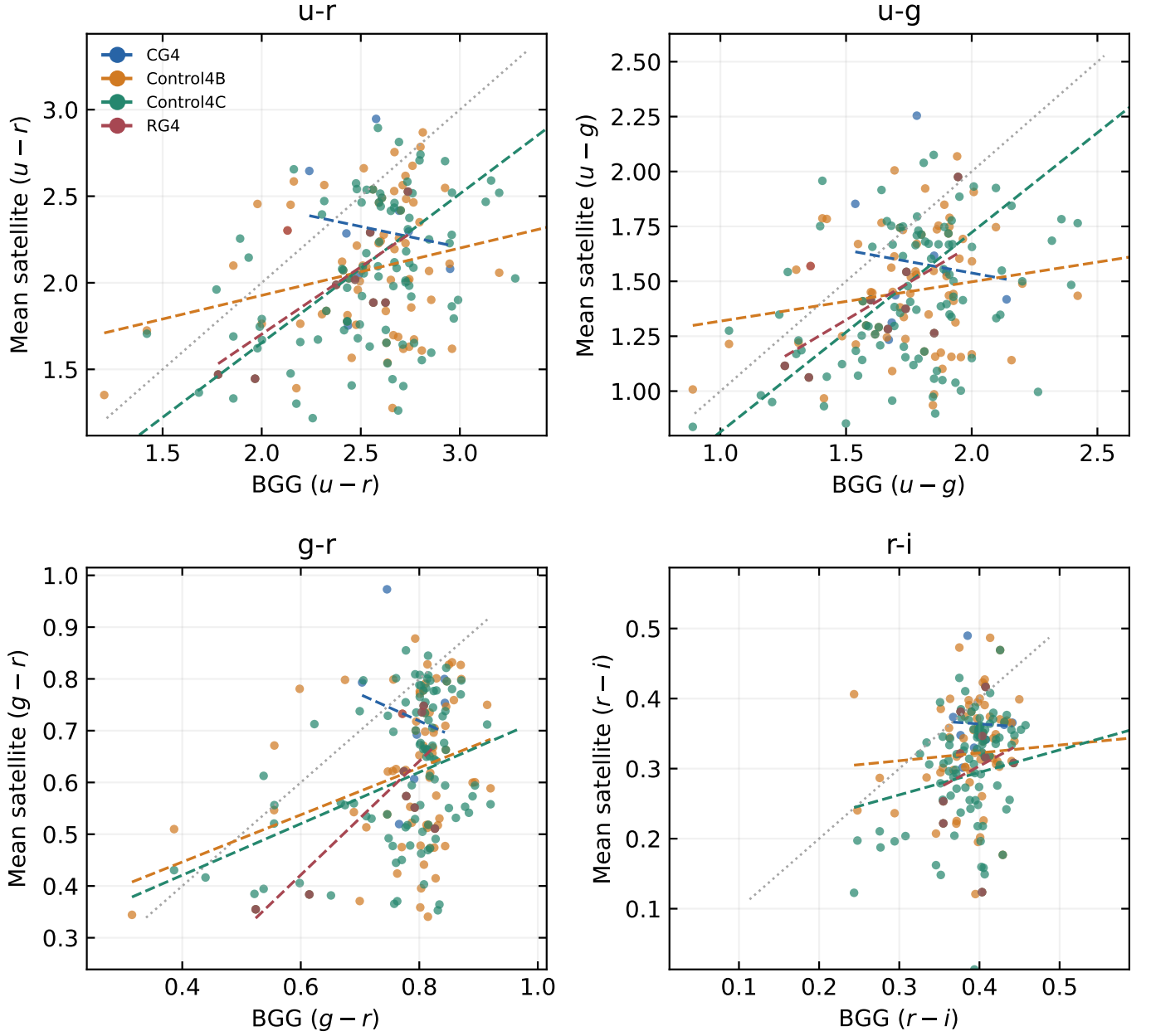


Fig. 22: Mean satellite colour versus BGG colour for groups with matched photometry. The dotted line is the one-to-one relation and dashed lines are catalogue-specific fits.

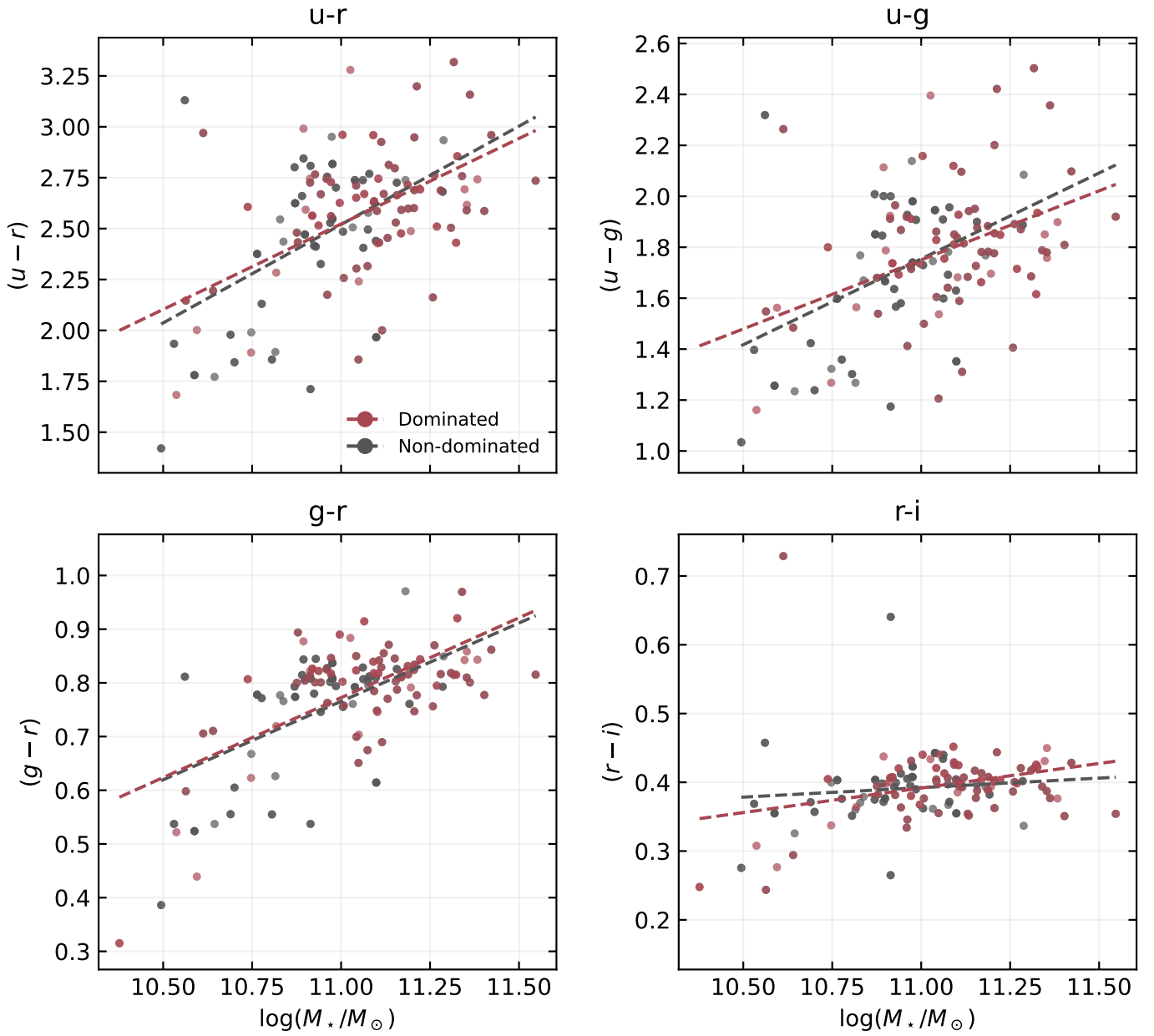


Fig. 23: BGG colour–mass relations split by luminosity domination. Comparing the fitted relations at a common stellar mass separates colour effects from the different mass distributions.

440 Interpolation points for sSFR classification

Acknowledgements. We thank ...

References

- Hahn, C., Starkenburg, T. K., Choi, E., et al. 2019, *Astrophys. J.*, 872
- Wuyts, S., Förster Schreiber, N. M., Lutz, D., et al. 2011, *Astrophys. J.*, 742, 96
- Zheng, Y.-L. & Shen, S.-Y. 2021, *ApJ*, 911, 105
- Bishop, C. M. 2006, *Pattern Recognition and Machine Learning* (Springer)
- Tojeiro, R., Heavens, A. F., Jimenez, R., & Panter, B. 2009, *Mon. Not. R. Astron. Soc.*, 397, 1263
- 450 Lintott, C., Schawinski, K., Bamford, S., et al. 2011, *MNRAS*, 410, 166
- Alpaslan, M., Driver, S., Robotham, A. S. G., et al. 2015, *MNRAS*, 451, 3249
- Kullback, S. & Leibler, R. A. 1951, *Ann. Math. Statist.*, 22, 79
- Veilleux, S. & Osterbrock, D. E. 1987, *Astrophysical Journal Supplement Series*, 63, 295
- Dempster, A. P., Laird, N. M., & Rubin, D. B. 1977, *J. R. Statist. Soc. B*, 39, 1
- 460 McLachlan, G. & Peel, D. 2000, *Finite Mixture Models* (Wiley)
- Tricottet, M., Mamon, G. A., & Díaz-Giménez, E. 2025, *A&A*, 699, A329
- Bisigello, L., Caputi, K. I., Grogin, N., & Koekemoer, A. 2018, *Astron. Astrophys.*, 609, A82
- Kauffmann, G., Heckman, T. M., Tremonti, C., et al. 2003, *Monthly Notices of the Royal Astronomical Society*, 346, 1055
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, *J. Mach. Learn. Res.*, 12, 2825
- Rousseeuw, P. J. 1987, *Comput. Appl. Math.*, 20, 53